

PROMPT ENGINEERING EXPERIMENTS

Application: AI Workspace Streamlit App

Instructions:

Run the app using: `streamlit run app.py`

Run each experiment in a new chat and paste the actual AI response under Output.

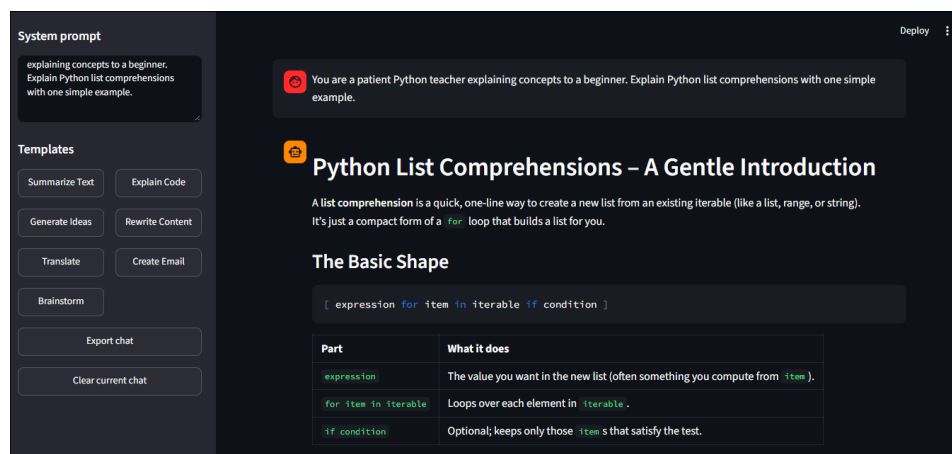
- **EXPERIMENT 1: ROLE PROMPTING**

Prompt:

You are a patient Python teacher explaining concepts to a beginner.

Explain Python list comprehensions with one simple example.

Output:



Observation:

The role made the response simpler, more beginner-friendly, and more instructional.

Improvement for the app:

Add role presets such as Teacher, Developer, Researcher, and Writer.

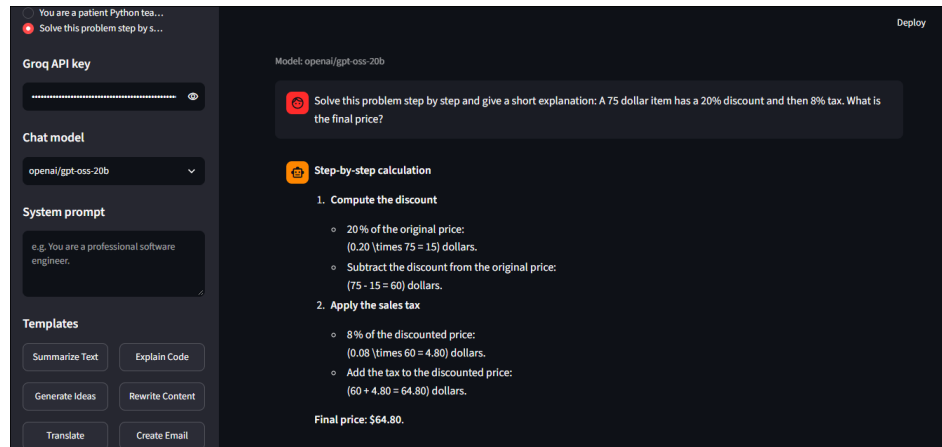
- **EXPERIMENT 2: CHAIN-OF-THOUGHT PROMPTING**

Prompt:

Solve this problem step by step and give a short explanation:

A 75 dollar item has a 20% discount and then 8% tax. What is the final price?

Output:



Observation:

Asking for steps made the calculation easier to understand and verify.

Improvement for the app:

Add response styles such as Answer Only, Brief Explanation, and Step-by-Step.

- **EXPERIMENT 3: FEW-SHOT PROMPTING**

Prompt:

Classify messages as Positive, Neutral, or Negative.

Example:

Message: The product arrived early.

Category: Positive

Example:

Message: The product is acceptable.

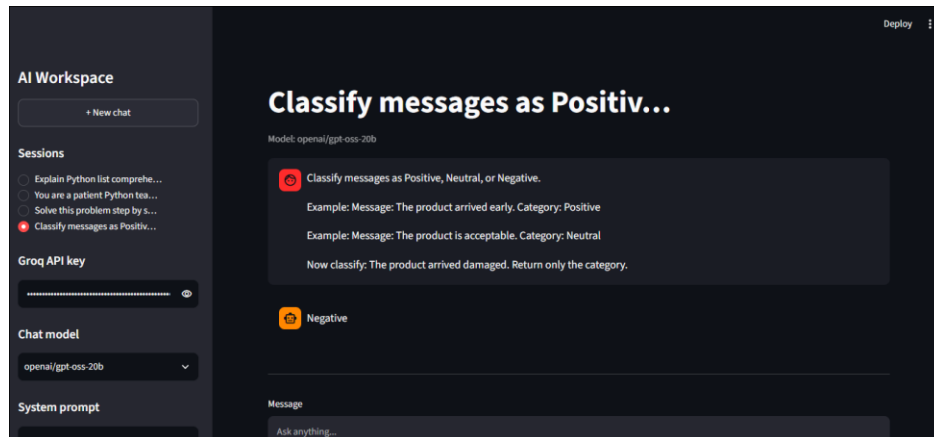
Category: Neutral

Now classify:

The product arrived damaged.

Return only the category.

Output:



Observation:

The examples helped the model understand the categories and output format.

Improvement for the app:

Add an Examples section where users can enter sample inputs and outputs.

• EXPERIMENT 4: STRUCTURED OUTPUT USING JSON

Prompt:

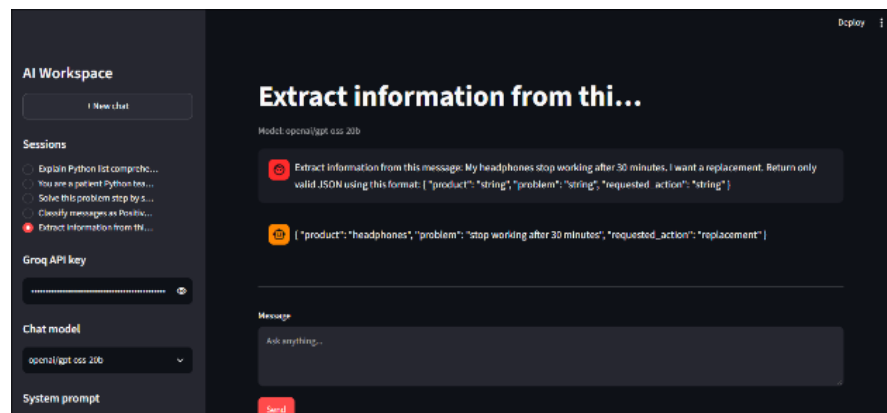
Extract information from this message:

My headphones stop working after 30 minutes. I want a replacement.

Return only valid JSON using this format:

```
{
  "product": "string",
  "problem": "string",
  "requested_action": "string"
}
```

Output:



Observation:

JSON makes the response predictable and easier for software to process.

Improvement for the app:

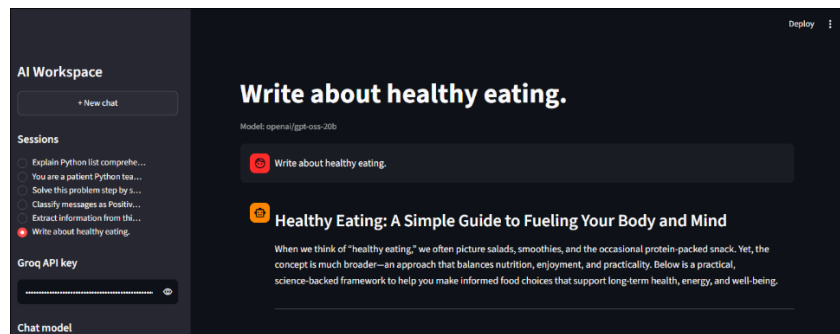
Add a JSON response option and validate the response using Python's `json.loads()`.

• EXPERIMENT 5: PROMPT OPTIMIZATION

Original prompt:

Write about healthy eating.

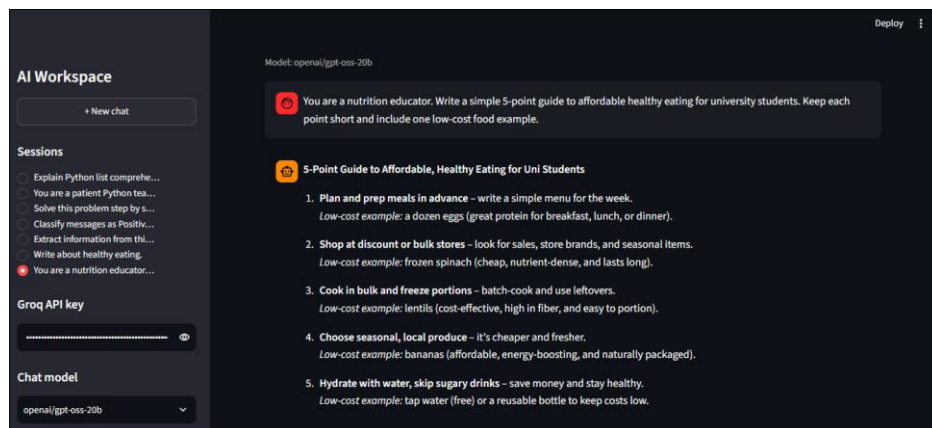
Output:



Improved prompt:

You are a nutrition educator. Write a simple 5-point guide to affordable healthy eating for university students. Keep each point short and include one low-cost food example.

Output:



Observation:

The improved prompt gave the model a role, audience, topic, length, and format.

The response was more focused and useful.

Improvement for the app:

Add a prompt optimizer that helps users add a role, audience, constraints, and output format.

CONCLUSION

These experiments show that better prompts produce clearer and more consistent AI responses. My AI Workspace could be improved with role presets, examples, JSON validation, response styles, and prompt comparison.